

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A food truck buys forks for ~~\$0.04~~ each and plates for ~~\$0.48~~ each. The total cost of x forks and y plates is ~~\$661.76~~. Which equation represents this situation?

Answer

- A. $0.48x - 0.04y = 661.76$
- B. $0.04x - 0.48y = 661.76$
- C. $0.48x + 0.04y = 661.76$
- D. $0.04x + 0.48y = 661.76$

Correct Answer: D

Rationale

Choice D is correct. It’s given that the food truck buys forks for ~~\$0.04~~ each. Therefore, the cost, in dollars, of x forks can be represented by the expression ~~0.04~~ x . It’s also given that the food truck buys plates for ~~\$0.48~~ each. Therefore, the cost, in dollars, of y plates can be represented by the expression ~~0.48~~ y . Since the total cost of x forks and y plates is ~~\$661.76~~, the equation ~~0.04~~ $x + 0.48y = 661.76$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation in which the food truck buys forks for ~~\$0.48~~ each and plates for ~~\$0.04~~ each.